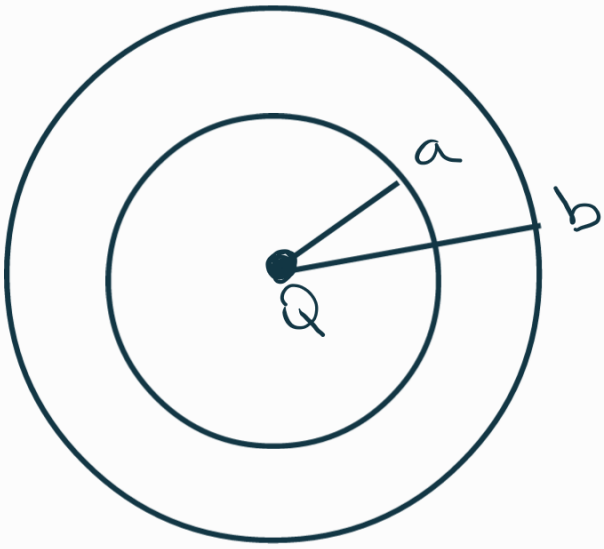
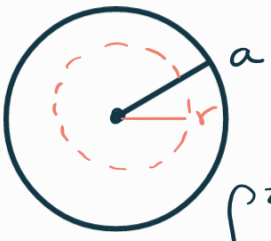


a) Magnitud del Campo en :

i) $r<a$



$$\oint \vec{E}\cdot d\vec{A} = \frac{q_{enc}}{\varepsilon_0}$$

$$\int_0^{2\pi}\int_0^{\pi} E r^2 \operatorname{sen}\phi \, d\phi \, d\theta = \frac{q_{enc}}{\varepsilon_0}$$

(1)

$$E r^2 \int_0^{2\pi} d\theta \int_0^{\pi} \operatorname{sen}\phi \, d\phi = \frac{q_{enc}}{\varepsilon_0}$$

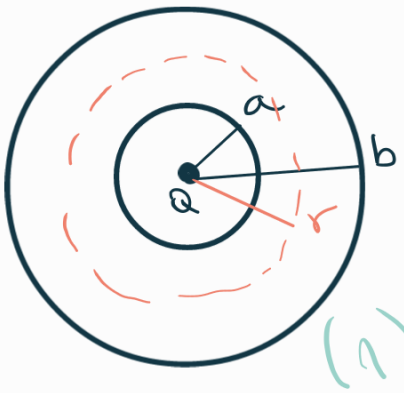
$$E r^2 (\theta)\Big|_0^{2\pi} (-\cos\phi)\Big|_0^{\pi} = \frac{q_{enc}}{\varepsilon_0}$$

$$E r^2 \, 2\pi \cdot (-2) = \frac{q_{enc}}{\varepsilon_0}$$

(2) $E\, 4\pi r^2 = \dfrac{q_{enc}}{\varepsilon_0}$ $\quad q_{enc} = Q$

$$\Rightarrow E = \frac{Q}{4\pi r^2 \varepsilon_0} \; [N/C]$$ (2)

ii) Para $a<r<b$

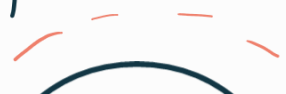


(1)

$E = 0 \; [N/C]$ (1)

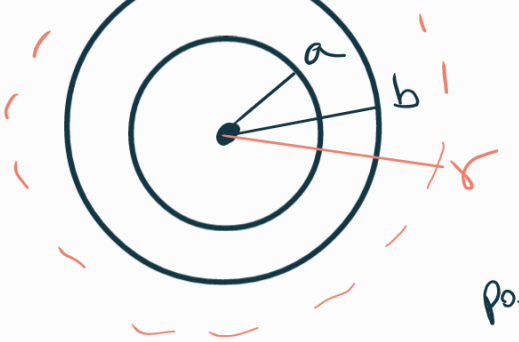
porque está dentro del Conductor.

iii)



Sabemos que :

$$\int_0^{2\pi}\int_0^{\pi} E r^2 4\pi$$



(1)

$$\oint \vec{E}\cdot d\vec{A} = \int_0 \int_0 E r^2 \operatorname{sen}\phi \, d\phi \, d\theta = E r^2 4\pi$$

(2)

y $q_{enc} = Q - 2Q = -Q$

por lo tanto

$$E r^2 4\pi = \frac{-Q}{\varepsilon_0}$$

$$E = \frac{Q}{r^2 4\pi \varepsilon_0} \; [N/C]$$

(2)

b) Para este caso, corresponde el campo de i)

$$\Rightarrow E = \frac{Q}{4\pi r^2 \varepsilon_0} \; [N/C] = \frac{110 \cdot 10^{-6} C}{4\pi (0{,}08 m)^2 \cdot 8{,}85 \cdot 10^{-12} \, C^2/Nm^2} = 1{,}54 \cdot 10^{8} \, [N/C]$$

(2)

(2)

c) Densidad volumétrica

$$\rho = \frac{Q}{V} = \frac{-2Q}{\frac{4}{3}\pi (b^3 - a^3)} = \frac{-2 \cdot 110 \cdot 10^{-6} C}{\frac{4}{3}\pi (0{,}15^3 - 0{,}1^3) m^3} = -0{,}22 \; [C/m^3]$$

(1)

(2)